

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 2

Year 12 Higher School Certificate
Trial Examination Term 3 2026

STUDENT NUMBER: _____ STUDENT NAME: _____ TEACHER: _____

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided separately
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination room.

Total marks – 100

Section I Pages 2 – 5

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet
provided

Section II Pages 6 – 11

90 marks

Attempt Questions 11 – 16

Start each question in a new writing booklet

Write your student number on every writing booklet

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/16	/15	/15	/15	/14	/15	/100

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment
Outcomes assessed: MEX12-1, MEX12-2, MEX12-3, MEX12-4, MEX12-5, MEX12-7, MEX12-8.

Section I

10 marks

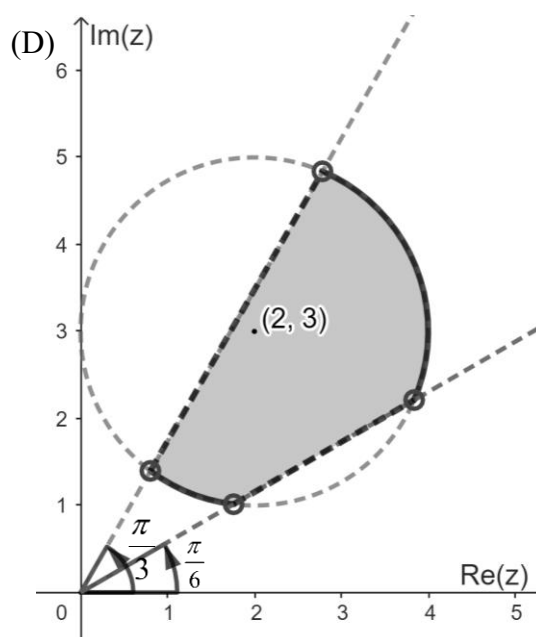
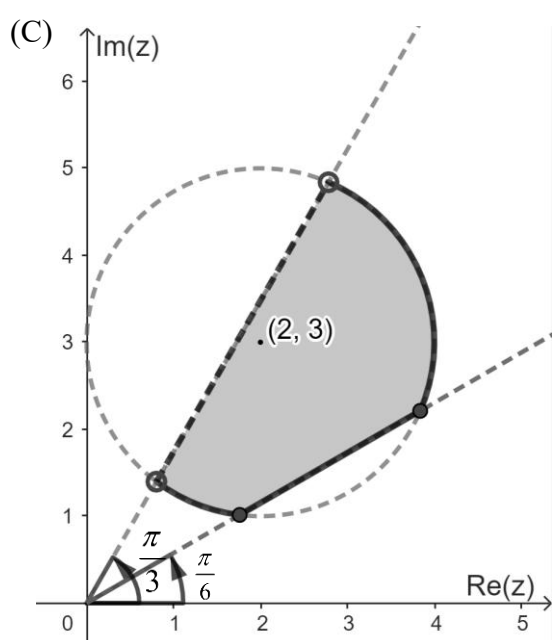
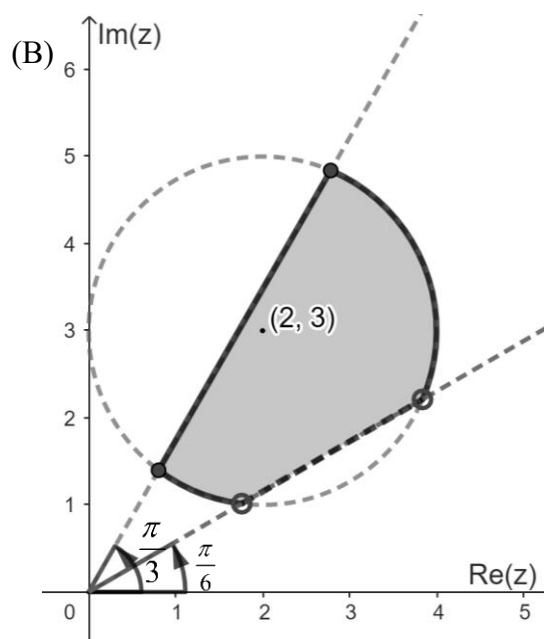
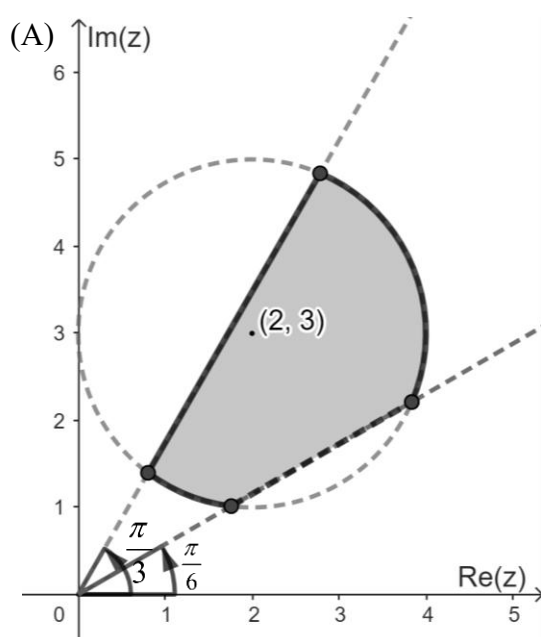
Attempt Questions 1 – 10

Allow about 15 minutes for this section.

Use the Objective Response answer sheet for Questions 1 – 10

1. Let R be the region in the complex plane defined by $|z - 2 - 3i| \leq 2$ and $\frac{\pi}{6} < \text{Arg}(z) \leq \frac{\pi}{3}$.

Which diagram best represents the shaded region?



2.

Consider the proposition.

‘If each term in a geometric series is smaller than the previous term, then $0 < r < 1$ ’.

Given the first term and common ratio in each of the following geometric series, which one disproves the proposition?

- (A) $T_1 = 2, \quad r = 2$
- (B) $T_1 = 8, \quad r = 0.5$
- (C) $T_1 = -2, \quad r = 2$
- (D) $T_1 = 8, \quad r = -0.5$

3. Consider the following statement.

‘If I have a scientific calculator in my bag and I study Extension 2 Mathematics, then I am a problem solver.’

Which of the following is the contrapositive of this statement?

- (A) If I am a problem solver, then I have a scientific calculator in my bag and I study Extension 2 Mathematics.
- (B) If I am not a problem solver, then I do not have a scientific calculator in my bag or I do not study Extension 2 Mathematics.
- (C) If I am a problem solver, then I do not have a scientific calculator in my bag or I do not study Extension 2 Mathematics.
- (D) If I am not a problem solver, then I do not have a scientific calculator in my bag but I study Extension 2 Mathematics.

4. Which of the following expressions would have an anti-derivative which includes $\sin^{-1}(f(x))$?

- (A) $\frac{2x}{\sqrt{x^4 - 2x^2 + 1}}$
- (B) $\frac{2x}{\sqrt{-x^4 - 4x^2 + 5}}$
- (C) $\frac{2x}{x^4 - 2x^2 + 1}$
- (D) $\frac{2x}{-x^4 - 4x^2 + 5}$

5. If $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{AB} = \underline{b}$ and $\overrightarrow{BO} = \underline{c}$, consider the statement:
“If $\underline{a} + \underline{b} + \underline{c} = 0$ then the points O, A and B form a triangle.”

The converse of the statement is:

- (A) If the points O, A and B do not form a triangle then $\underline{a} + \underline{b} + \underline{c} \neq 0$.
- (B) If $\underline{a} + \underline{b} + \underline{c} \neq 0$ then O, A and B do not form a triangle.
- (C) If the points O, A and B form a triangle then $\underline{a} + \underline{b} + \underline{c} \neq 0$.
- (D) If the points O, A and B form a triangle then $\underline{a} + \underline{b} + \underline{c} = 0$.
6. A particle of mass 100 kilograms is moving in three-dimensional space, and is subject to a constant force.

Its displacement in metres from a fixed point after t seconds is

$$\underline{r} = \begin{pmatrix} 4t^2 + 4t + 1 \\ 3t - 4 \\ 5 - 3t^2 \end{pmatrix}.$$

What is the magnitude of the force acting on the particle?

- (A) 200 N
- (B) 648 N
- (C) 1000 N
- (D) 1374 N
7. A particle moving in a straight line obeys $v^2 = 9(5 + 4x - x^2)$ where x is the displacement from the origin in metres and v is its velocity in ms^{-1} . Where is the centre of the motion?
- (A) $x = -1$
- (B) $x = 0$
- (C) $x = 2$
- (D) $x = 5$

8. A stone is thrown horizontally with speed $V = \sqrt{2gH}$ ms⁻¹ from the top point O of a vertical tower of height H metres, standing on horizontal ground. It moves in a vertical plane under gravity where the acceleration due to gravity is g ms⁻². At time t seconds the stone has position vector $Vt\hat{i} - \frac{1}{2}gt^2\hat{j}$ relative to O . The stone hits the ground at time T seconds at a distance X metres from the foot of the tower. What is the value of X ?

(A) $\frac{1}{2}H$

(B) H

(C) $2H$

(D) $4H$

9.

Let $g(x)$ be a function with first derivative given by $g'(x) = \int_0^x e^{-t^2} dt$.

Which of the following must be true on the interval $0 < x < 2$?

(A) $g(x)$ is increasing and the graph of $g(x)$ is concave up.

(B) $g(x)$ is increasing and the graph of $g(x)$ is concave down.

(C) $g(x)$ is decreasing and the graph of $g(x)$ is concave up.

(D) $g(x)$ is decreasing and the graph of $g(x)$ is concave down.

10.

The complex number $z = x + iy$ satisfies $|z| \leq y + 1$ and $|z| \leq x + 1$.

What is the maximum possible value of $|z|$?

(A) $1 - \sqrt{2}$

(B) $1 + \sqrt{2}$

(C) $2 - \sqrt{2}$

(D) $2 + \sqrt{2}$

END OF SECTION I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section.

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) **START A NEW WRITING BOOKLET**

- (a) Express $\frac{4-3i}{3+2i}$ in the form $x+iy$, where x and y are real numbers. 2
- (b) Evaluate $\int \cos^3 x \sin^2 x \, dx$. 2
- (c) (i) Write the complex number $-1+\sqrt{3}i$ in exponential form. 2
- (ii) Hence, find the exact value of $(-1+\sqrt{3}i)^8$, giving your answer in the form $x+iy$. 2
- (d) A triangle PQR is formed in three-dimensional space with vertices $P(3, -2, 4)$, $Q(-2, 1, 5)$ and $R(5, -1, 2)$. Find the size of $\angle QRP$, giving your answer to the nearest degree. 3
- (e) Let $\mathbf{r}_1 = \begin{bmatrix} 5 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\lambda \in \mathbb{R}$, be the vector equation of the line L_1 . A second line L_2 passes through the point $P(-2, 4)$ and is parallel to L_1 . Find the equation of the line L_2 in the form $y = mx + c$. 2
- (f) Using the substitution $t = \tan \frac{x}{2}$, find $\int \frac{dx}{1 - \sin x}$. 3

End of Question 11

Question 12 (15 marks) START A NEW WRITING BOOKLET

(a) (i) For positive real numbers a and b , prove that $\frac{a+b}{2} \geq \sqrt{ab}$. **2**

(ii) Given $a \geq b > 0$, prove that $\frac{1}{a} \leq \frac{1}{b}$. **1**

(iii) Given positive real numbers p and q , consider $f(x) = \frac{x}{(x+p)(x+q)}$.

By using parts (i) and (ii), or otherwise, prove that for $x > 0$ **3**

$$f(x) \leq \frac{1}{(\sqrt{p} + \sqrt{q})^2}.$$

(b) (i) Find integers A , B and C such that:

$$\frac{x^2 + 3x + 18}{(x^2 + 4)(x - 5)} = \frac{Ax + B}{x^2 + 4} + \frac{C}{x - 5} \quad \text{3}$$

(ii) Hence find $\int \frac{x^2 + 3x + 18}{(x^2 + 4)(x - 5)} dx$ **2**

(c) A particle moving along the x axis. At time t seconds it has displacement x metres from the origin O , velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$ given by $a = -\frac{1}{50}(10 - x)^3$. Initially the particle is at O and moving to the right of O with speed 10 ms^{-1} .

(i) Use integration to show that $v = \frac{1}{10}(10 - x)^2$. **2**

(ii) Hence use integration to show that $x = \frac{10t}{1+t}$. **2**

End of Question 12

Question 13 (15 marks) **START A NEW WRITING BOOKLET**

(a) Prove by contradiction that $\log_4 7$ is irrational. **3**

(b) A sequence is defined recursively as $u_0 = 5$, $u_n = 2u_{n-1} + 1$ for $n \geq 1$. **4**

Use mathematical induction to prove that $u_n = 3(2^{n+1}) - 1$ for all positive integers n .

(c) (i) Solve the equation $z^5 = 1$, giving the roots in modulus/argument form. **2**

(ii) Hence show that $z^4 + z^5 + z^2 + z + 1 = \left(z^2 - 2z \cos \frac{2\pi}{5} + 1\right) \left(z^2 - 2z \cos \frac{4\pi}{5} + 1\right)$. **3**

(iii) Show that $4 \left(1 - \cos \frac{2\pi}{5}\right) \left(1 - \cos \frac{4\pi}{5}\right) = 5$. **1**

(iv) Hence show that $x = \cos \frac{2\pi}{5}$ is a root of the equation $8x^3 - 8x^2 - 8x + 3 = 0$. **2**

End of Question 13

Question 14 (15 marks) START A NEW WRITING BOOKLET

- (a) The point of intersection of the line through $A \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $B \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix}$ with the line through $C \begin{pmatrix} 0 \\ 9 \\ 10 \end{pmatrix}$ and $D \begin{pmatrix} 1 \\ 7 \\ 7 \end{pmatrix}$ is E .

(i) Prove that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$. 1

(ii) Find the coordinates of E . 2

- (b) Given that $z = 1 - \sqrt{3}i$ find the two square roots of z in the form $a + ib$. 3

- (c) Consider the complex number ω , which is the non-real solution to $z^5 - 1 = 0$ with the smallest principal argument. 3

Using the fact that $z = \omega^2 + \frac{1}{\omega^2}$ is a root of $z^2 + z - 1 = 0$, show that

$$\cos \frac{4\pi}{5} = -\frac{\sqrt{5} + 1}{4}.$$

- (d) Let points A and B on the complex plane represent the complex numbers $z_A = 1 + i$ and $z_B = -4$ respectively.

The isosceles triangle ABC is right-angled at C .

(i) Find the possible values of z_C , represented by the point C . 4

(ii) The line between, but not including the points A and B , has equation 2

$$\arg(z + x_1 + iy_1) - \arg(z + x_2 + iy_2) = \theta.$$

Find the values of x_1, y_1, x_2, y_2 and θ .

End of Question 14

Question 15 (14 marks) START A NEW WRITING BOOKLET

- (a) A particle is undergoing simple harmonic motion on the x -axis about the origin. It is initially at its extreme positive position. The amplitude of the motion is 10 and the particle returns to its initial position every 7 seconds.
- (i) Write an equation for the position of the particle at time t seconds. **1**
- (ii) How long does the particle take to move from a rest position to the point halfway between the rest position and the equilibrium position? **1**
- (b) A particle of mass m kg is set in motion with speed u ms⁻¹ and moves in a straight line before coming to rest. After time t seconds the particle has displacement x metres from its starting point O , velocity v ms⁻¹ and acceleration a ms⁻². The resultant force acting on the particle directly opposes its motion and has a magnitude $m(1 + v)$ Newtons so that $a = -(1 + v)$.
- (i) Find expressions for x in terms of v , v in terms of t and x in terms of t . **6**
- (ii) Show that $x + v + t = u$. **1**
- (iii) Find the distance travelled and the time taken by the particle in coming to rest. **2**
- (c) Use integration by parts to evaluate $\int_0^1 \sin^{-1} x \, dx$. **3**

End of Question 15

Question 16 (15 marks) START A NEW WRITING BOOKLET

(a) If $a > 1, b > 1$ and $c > 1$ such that $\frac{a}{b} \geq \frac{c}{a}$, show that $\frac{\ln a}{\ln b} \geq \frac{\ln c}{\ln a}$. **4**

(b) Let $I_n = \int_1^e (\ln x)^n dx$ for $n = 0, 1, 2, \dots$

(i) Prove that $I_{n+1} = e - (n+1)I_n$ for $n = 0, 1, 2, \dots$ and hence prove $I_2 = e - 2$. **3**

(ii) Use mathematical induction to prove that $I_n = (-1)^{n+1} n! + (-1)^n e n! \sum_{r=2}^n \frac{(-1)^r}{r!}$ **3**
for all positive integers $n \geq 2$.

(c) Given three vectors $\vec{a}, \vec{b}$, and $\vec{c}$, the vectors $\vec{u}, \vec{v}, \vec{w}$ are defined as

$$\vec{u} = (\vec{b} \cdot \vec{c}) \vec{a} - (\vec{c} \cdot \vec{a}) \vec{b}$$

$$\vec{v} = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

$$\vec{w} = (\vec{b} \cdot \vec{a}) \vec{c} - (\vec{c} \cdot \vec{b}) \vec{a}$$

The vectors $\vec{a}, \vec{b}, \vec{c}$ form a triangle.

(i) Prove $\vec{u}, \vec{v}, \vec{w}$ also form a triangle. **2**

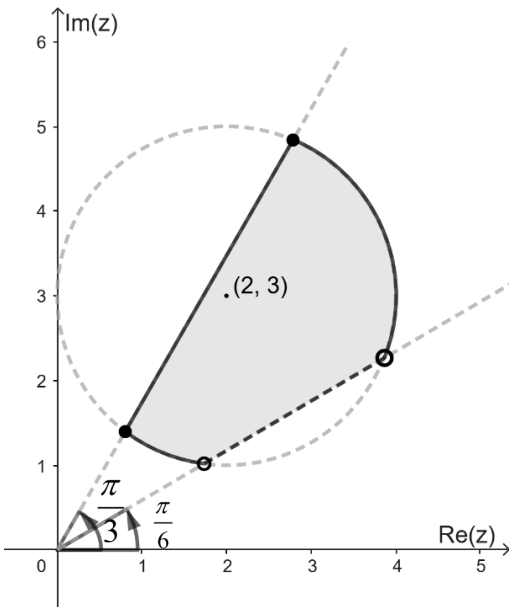
(ii) Calculate $\vec{u} \cdot \vec{c}$. **1**

(iii) Prove that the two triangles formed by $\vec{a}, \vec{b}, \vec{c}$ and $\vec{u}, \vec{v}, \vec{w}$ are similar. **2**

END OF ASSESSMENT

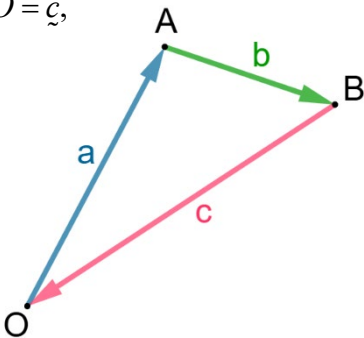
Section I Multiple Choice

Question	Answers
1	B
2	C
3	B
4	B
5	D
6	C
7	C
8	C
9	A
10	D

	Suggested Solution	Markers' Comments
1.	$z - 2 - 3i \leq 2$ and $\frac{\pi}{6} < \text{Arg}(z) \leq \frac{\pi}{3}$ Circle centre (2, 3) Radius = 2 units including points on circumference. A region between the two rays including points on the ray $y = \sqrt{3}x$, but excluding points on $y = \frac{1}{\sqrt{3}}x$ Intersecting points of circle and rays are included along $y = \sqrt{3}x$ but excluded on the ray $y = \frac{1}{\sqrt{3}}x$.  Answer B.	92.5% students correct.
2.	The proposition is in the form $P \Rightarrow Q$, where: P: each term in a geometric series is smaller than the previous term Q: $0 < r < 1$.	57.5% students correct.

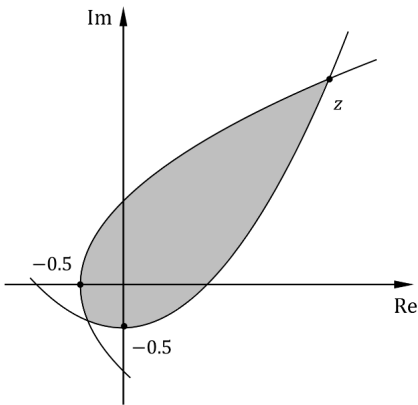
	Suggested Solution	Markers' Comments
2.	Continues: To disprove the proposition we need to find P and not Q occurring simultaneously, so each term in a geometric series is smaller than the previous term AND r is not between 0 and 1. Writing out the first three terms in each series: A: 2, 4, 8, which is not in descending order. Eliminate A. B: 8, 4, 2, which is in descending order but has $0 < r < 1$. Eliminate B. C: -2, -4, -8, which is in descending order and where r is not between 0 and 1. Correct. D: 8, -4, 2, which is not in descending order. Eliminate D. Answer C. <i>Alternative solution :</i> The proposition is in the form $P \Rightarrow Q$ where: P: each term in a geometric series is smaller than the previous term Q: $0 < r < 1$. $\neg(P \Rightarrow Q) \Leftrightarrow (P \wedge \neg Q)$. Writing out the first three terms in each series: A: 2, 4, 8, $(\neg P \wedge \neg Q)$. Eliminate A. B: 8, 4, 2, $(P \wedge Q)$. Eliminate B. C: -2, -4, -8, $(P \wedge \neg Q)$. Correct. D: 8, -4, 2, $(\neg P \wedge \neg Q)$. Eliminate D. Answer C.	
3.	Original statement: If I have a scientific calculator in my bag and I study Extension 2 Mathematics, then I am a problem solver. i.e. $(P \cap Q) \Rightarrow R$ $P = \{ \text{I have a scientific calculator in my bag} \}$ $Q = \{ \text{I study Extension 2 Mathematics} \}$ $R = \{ \text{I am a problem solver} \}$ Contraposition: $\neg R \Rightarrow \neg(P \cap Q)$ $\neg R \Rightarrow \neg P \cup \neg Q$ $\neg R \Rightarrow \neg P \text{ or } \neg Q$ If I am not a problem solve, then I do not have a scientific calculator in my bag or I do not study Extension 2 Mathematics. (B)	95% students correct.

	Suggested Solution	Markers' Comments
4.	The integrand has to be able to be written in the form $\frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}$ A: $\frac{2x}{\sqrt{x^4 - 2x^2 + 1}} = \frac{2x}{\sqrt{(x^2 - 1)^2}}$ $= \frac{\frac{d}{dx}(x^2 - 1)}{ x^2 - 1 }$ INCORRECT. FYI: The anti-derivative is $\frac{(x^2 - 1) \ln x^2 - 1 }{ x^2 - 1 } + c.$ B: $\frac{2x}{\sqrt{-x^4 - 4x^2 + 5}}$ $= \frac{\frac{d}{dx}(x^2 + 2)}{\sqrt{3^2 - (x^2 + 2)^2}}$ CORRECT. FYI: The anti-derivative is $\sin^{-1}\left(\frac{x^2 + 2}{3}\right) + c.$ C: $\frac{2x}{x^4 - 2x^2 + 1}$ $= \frac{\frac{d}{dx}(x^2 - 1)}{(x^2 - 1)^2}$ INCORRECT. FYI: The anti-derivative is $-\frac{1}{x^2 - 1} + c.$ D: $\frac{2x}{-x^4 - 4x^2 + 5}$ $= \frac{\frac{d}{dx}(x^2 + 2)}{3^2 - (x^2 + 2)^2}$ INCORRECT. FYI: The anti-derivative is $\frac{\ln(x^2 + 5) - \ln x^2 - 1 }{6} + c.$ Answer B. Alternative solution I: The integrand has to be able to be written in the form $\frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}$ The denominator has to be the square root of the difference of a constant and the square of a function of x, and the numerator has to be the derivative of the function of x.	

	Suggested Solution	Markers' Comments
4.	Continues A: $\frac{2x}{\sqrt{x^4 - 2x^2 + 1}} = \frac{2x}{\sqrt{(x^2 - 1)^2}}$ Eliminate A since the denominator is not the square root of the difference of a constant and the square of a function of x. B: $\frac{2x}{\sqrt{-x^4 - 4x^2 + 5}} = \frac{\frac{d}{dx}(x^2 + 2)}{\sqrt{3^2 - (x^2 + 2)^2}}$ CORRECT, since the denominator is the square root of the difference of a constant and the square of a function of x, and the numerator is the derivative of the function of x. C: $\frac{2x}{x^4 - 2x^2 + 1} = \frac{\frac{d}{dx}(x^2 - 1)}{(x^2 - 1)^2}$ Eliminate C since the denominator is not the square root of the difference of a constant and the square of a function of x. D: $\frac{2x}{-x^4 - 4x^2 + 5} = \frac{\frac{d}{dx}(x^2 + 2)}{3^2 - (x^2 + 2)^2}$ Eliminate D since the denominator is not the square root of the difference of a constant and the square of a function of x. Answer B.	
5.	If $\vec{OA} = \underline{a}$, $\vec{AB} = \underline{b}$ and $\vec{BO} = \underline{c}$, Original statement: "If $\underline{a} + \underline{b} + \underline{c} = 0$ then the points O, A and B form a triangle." i.e. $P = \{ \underline{a} + \underline{b} + \underline{c} = 0 \}$ $Q = \{ \text{the points } O, A \text{ and } B \text{ form a triangle} \}$ Converse statement: If Q, then P If the points O, A and B form a triangle, then $\underline{a} + \underline{b} + \underline{c} = 0$ Answer D. 	92.5% students correct.

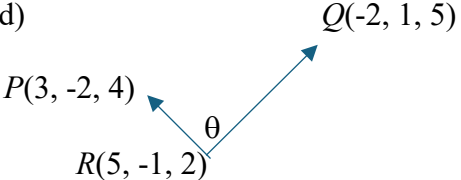
	Suggested Solution	Markers' Comments
6.	$\underset{\sim}{r} = \begin{pmatrix} 4t^2 + 4t + 1 \\ 3t - 4 \\ 5 - 3t^2 \end{pmatrix}$ $\underset{\sim}{v} = \begin{pmatrix} 8t + 4 \\ 3 \\ -6t \end{pmatrix}$ $\underset{\sim}{a} = \begin{pmatrix} 8 \\ 0 \\ -6 \end{pmatrix}$ $ \underset{\sim}{a} = \sqrt{8^2 + 0^2 + 6^2}$ $= 10 \text{ ms}^{-2}$ $F = m \times a$ $= 100 \times 10$ $= 1000 \text{ N}$ Answer C.	92.5% students correct.
7.	$v^2 = 9(5 + 4x - x^2)$ $v^2 = -9(x^2 - 4x - 5)$ $v^2 = -9(x^2 - 4x + 4 - 5 - 4)$ $v^2 = -9((x - 2)^2 - 9)$ $v^2 = 9(9 - (x - 2)^2)$ Hence centre is at $x = 2$ Answer C.	95% students correct.
8.	$\frac{1}{2}gT^2 = H \text{ and } X = VT = \sqrt{2gHT}$ $\Rightarrow \frac{2H}{g} = \frac{X^2}{2gH}$ $\therefore X = 2H$ Answer C.	Poorly done (40% students correct.)
9.	$g'(x) = \int_0^x e^{-t^2} dt.$ As $e^{-t^2} > 0$, $0 \leq t < 2$, then $g'(x) = \int_0^x e^{-t^2} dt > 0$ i.e. $g(x)$ is increasing. $g''(x) = \frac{d}{dx} \int_0^x e^{-t^2} dt = e^{-x^2} > 0, 0 \leq t < 2,$ i.e. $g(x)$ is concave up. Answer A	Poorly done (27.5% students correct.) Many students did not consider that the exponential function $e^{-t^2} > 0$ for all real t values which makes both $g'(x)$ and $g''(x)$ positive.

	Suggested Solution	Markers' Comments
10.	The inequalities are symmetrical in x and y, so the maximum occurs when $y = x$. $ z = \sqrt{x^2 + x^2}$ $= \sqrt{2x^2}$ $\therefore \sqrt{2x^2} = x + 1$ $2x^2 = x^2 + 2x + 1$ $x^2 - 2x - 1 = 0$ $x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)}$ $= \frac{2 \pm 2\sqrt{2}}{2}$ $= 1 \pm \sqrt{2}$ $y = 1 \pm \sqrt{2}$ $z _{\max}$ when $x = y = 1 + \sqrt{2}$ since $1 + \sqrt{2} > 1 - \sqrt{2}$ $z _{\max} = \sqrt{2}(1 + \sqrt{2})$ using the 45° exact triangle $= \sqrt{2} + 2$ Answer D. Alternative solution I: $ z \leq y + 1 \quad (1)$ $ z \leq x + 1 \quad (2)$ $(2) - (1): 0 \leq x - y \rightarrow y \leq x \quad (3)$ $(1) - (2): 0 \leq y - x \rightarrow y \geq x \quad (4)$ From (3) and (4): $y = x \quad (5)$ $ z = \sqrt{x^2 + y^2}$ From (5): $z = \sqrt{x^2 + x^2}$ $= \sqrt{2x^2}$ sub in (2): $\therefore \sqrt{2x^2} = x + 1$ $2x^2 = x^2 + 2x + 1$ $x^2 - 2x - 1 = 0$ $x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)}$ $= \frac{2 \pm 2\sqrt{2}}{2}$ $= 1 \pm \sqrt{2}$ $y = 1 \pm \sqrt{2}$ $z _{\max}$ when $x = y = 1 + \sqrt{2}$ since $1 + \sqrt{2} > 1 - \sqrt{2}$ $z _{\max} = \sqrt{2}(1 + \sqrt{2})$ using the 45° exact triangle $= \sqrt{2} + 2$	Poorly done (32.5% students correct.) Many students (52,5%) chose (B) not realising that it is the coordinates of z and not the modulus of z.

	Suggested Solution	Markers' Comments
10.	Answer D. Alternative solution II: $ z \leq x + 1$ $\sqrt{x^2 + y^2} \leq x + 1$ $x^2 + y^2 \leq x^2 + 2x + 1$ $y^2 \leq 2x + 1$ $2x \geq y^2 - 1$ $x \geq \frac{y^2 - 1}{2}$ Which is a concave right parabola with vertex $(-\frac{1}{2}, 0)$. Similarly $ z \leq y + 1 \rightarrow y \geq \frac{x^2 - 1}{2}$ which is a concave up parabola with vertex $(0, -\frac{1}{2})$.  The region is graphed at right, with z a maximum at the top right. Using symmetry, the maximum value of z occurs at the point of intersection in the first quadrant where $y = x$. $\therefore x = \frac{x^2 - 1}{2}$ $2x = x^2 - 1$ $x^2 - 2x - 1 = 0$ $x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)}$ $= \frac{2 \pm 2\sqrt{2}}{2}$ $= 1 \pm \sqrt{2}$ $y = 1 \pm \sqrt{2}$ $ z _{\max} = \sqrt{2}(1 + \sqrt{2}) \quad \text{using the } 45^\circ \text{ exact triangle}$ $= \sqrt{2} + 2$ Answer D.	

SECTION 11

11.	(a) $\frac{4-3i}{3+2i} = \frac{4-3i}{3+2i} \times \frac{3-2i}{3-2i}$ $= \frac{(4-3i)(3-2i)}{9+4}$ $= \frac{12-17i-6}{13}$ $= \frac{6}{13} - \frac{17}{13}i$ ①	Mostly well done. (87.5% correct)
11.	(b) $\int \cos^3 x \sin^2 x \, dx = \int \cos x \cos^2 x \sin^2 x \, dx$ $= \int \cos x (1 - \sin^2 x) \sin^2 x \, dx$ $= \int \cos x (\sin^2 x - \sin^4 x) \, dx$ $= \int \cos x \sin^2 x - \cos x \sin^4 x \, dx$ $= \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$ ①	Mostly well done (95% students correct.)
11.	(c)(i) $-1 + \sqrt{3}i = \sqrt{(-1)^2 + (\sqrt{3})^2}$ $= 2$ $\text{Arg}(-1 + \sqrt{3}i) = \tan^{-1} \frac{\sqrt{3}}{-1}$ (QII) $= \pi - \frac{\pi}{3}$ $= \frac{2\pi}{3}$ ① $\therefore -1 + \sqrt{3}i = 2e^{\frac{2\pi}{3}i}$ ① (c)(ii) $(-1 + \sqrt{3}i)^8 = \left(2e^{\frac{2\pi}{3}i}\right)^8$ or $\left(2\text{cis}\frac{2\pi}{3}\right)^8$ $= 2^8 e^{\frac{16\pi}{3}i} = 2^8 \text{cis}\left(8\left(\frac{2\pi}{3}\right)\right)$ $= 256e^{(5\pi + \frac{\pi}{3})i} = 256 \text{cis}\frac{16\pi}{3}$ (QIII) ① $= 256\left(-\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$ $= 256\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$ $= -128 - 128\sqrt{3}i$ ①	Mostly well done. (87.5% correct) Area of concern: (1) students not realising $\frac{16\pi}{3}$ is in QIII.

11.	(d)  $\overrightarrow{RP} = \overrightarrow{OP} - \overrightarrow{OR}$ and $\overrightarrow{RQ} = \overrightarrow{OQ} - \overrightarrow{OR}$ $= \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} - \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix} - \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix}$ $= \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} \quad \text{①} \qquad = \begin{bmatrix} -7 \\ 2 \\ 3 \end{bmatrix} \quad \text{①}$ $\cos \theta = \frac{\overrightarrow{RP} \bullet \overrightarrow{RQ}}{ \overrightarrow{RP} \overrightarrow{RQ} }$ $= \frac{14 - 2 + 6}{\sqrt{(-2)^2 + (-1)^2 + (2)^2} \times \sqrt{(-7)^2 + (2)^2 + (3)^2}}$ $= \frac{18}{\sqrt{9} \times \sqrt{62}}$ $= \frac{6}{\sqrt{62}}$ $\therefore \theta = 40.36^\circ$ $\therefore \theta = 40^\circ$ (to nearest degree) ①	Mostly well done. (75% correct)
11.	(e) $L_1: \quad \vec{r}_1 = \begin{bmatrix} 5 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \lambda \in \mathbb{R}.$ Given: $L_2 \parallel L_1, \quad \therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{①}$ i.e. $x = -2 + 2\mu \quad \text{--- ①}$ $y = 4 + 3\mu \quad \text{--- ②}$ From ① $2\mu = x + 2$ $\mu = \frac{x}{2} + 1 \quad \text{--- ①'}$ Sub ①' into ② $y = 4 + 3\left(\frac{x}{2} + 1\right)$ $y = 4 + \frac{3x}{2} + 3$ $\therefore y = \frac{3}{2}x + 7 \quad \text{①}$	Mostly well done. (85% correct)

11.	(f) $\int \frac{dx}{1-\sin x}$ where $t = \tan \frac{x}{2}$ and $\sin x = \frac{2t}{1+t^2}$ $\frac{x}{2} = \tan^{-1} t$ $x = 2 \tan^{-1} t$ $\frac{dx}{dt} = \frac{2}{1+t^2}$ $dx = \frac{2}{1+t^2} dt$ $= \int \frac{\frac{2}{1+t^2}}{1 - \frac{2t}{1+t^2}} dt \quad \textcircled{1}$ $= \int \frac{\frac{2}{1+t^2}}{\frac{1+t^2-2t}{1+t^2}} dt$ $= \int \frac{2}{t^2 - 2t + 1} dt$ $= 2 \int \frac{1}{(t-1)^2} dt \quad \textcircled{1}$ $= 2 \int (t-1)^{-2} dt$ $= 2 \left[\frac{(t-1)^{-1}}{-1} \right] + C$ $= -\frac{2}{t-1} + C$ $= -\frac{2}{\tan \frac{x}{2} - 1} + C \quad \textcircled{1}$	Mostly well done. (67.5% correct) Area of concern: (1) students did not answer in terms of x.
12.	(a)(i) $(\sqrt{a} - \sqrt{b})^2 \geq 0$ $\Rightarrow a - 2\sqrt{ab} + b \geq 0$ $\Rightarrow a + b \geq 2\sqrt{ab}$ $\Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$ (a)(ii) $\frac{1}{a} - \frac{1}{b} = \frac{b-a}{ab}$ $\leq 0, \text{ since } a \geq b \Rightarrow b-a \leq 0$ Hence, $\frac{1}{a} \leq \frac{1}{b}$	

12.	(a)(iii) $f(x) = \frac{x}{(x+p)(x+q)}$ $\frac{1}{f(x)} = \frac{(x+p)(x+q)}{x}$ $= \frac{x^2 + px + qx + pq}{x}$ $\frac{1}{f(x)} = x + (p+q) + \frac{pq}{x}$ $\frac{1}{f(x)} = (p+q) + x + \frac{pq}{x}$ But $\frac{a+b}{2} \geq \sqrt{ab}$ from (i) $a+b \geq 2\sqrt{ab}$ So if $a = x$ and $b = \frac{pq}{x}$ $x + \frac{pq}{x} \geq 2\sqrt{x \times \frac{pq}{x}}$ $x + \frac{pq}{x} \geq 2\sqrt{pq}$ $\frac{1}{f(x)} \geq p + q + 2\sqrt{pq}$ $\frac{1}{f(x)} \geq p + 2\sqrt{pq} + q$ $\frac{1}{f(x)} \geq (\sqrt{p})^2 + 2\sqrt{pq} + (\sqrt{q})^2$ $\frac{1}{f(x)} \geq (\sqrt{p} + \sqrt{q})^2 \geq 0$ Hence: $f(x) \leq \frac{1}{(\sqrt{p} + \sqrt{q})^2}$, using part (ii)	
12.	(b) (i) $\frac{x^2+3x+18}{(x^2+4)(x-5)} = \frac{Ax+B}{x^2+4} + \frac{C}{x-5}$ $(x^2 + 3x + 18) = (Ax + B)(x - 5) + C(x^2 + 4)$ Let $x = 5 \Rightarrow 58 = 29C$ $C = \frac{58}{29} = 2$ Let $x = 0 \Rightarrow 18 = -5B + 4(2)$ $-5B = 10$ $B = -2$ Let $x = 1$	

	$22 = (A - 2)(-4) + 2(5)$ $22 = -4A + 8 + 10$ $4 = -4A$ $A = -1$ $A = -1, B = -2 \text{ and } C = 2$ $(ii) \int \frac{x^2+3x+18}{(x^2+4)(x-5)} dx$ $= \int \frac{-x-2}{x^2+4} + \frac{2}{x-5} dx \quad (\text{From part (i)})$ $= \int -\frac{x}{x^2+4} dx + \int -\frac{2}{x^2+4} dx + \int \frac{2}{x-5} dx$ $= -\frac{1}{2} \int \frac{2x}{x^2+4} dx - 2 \int \frac{1}{x^2+4} dx + \int \frac{2}{x-5} dx$ $= -\frac{1}{2} \ln x^2+4 - 2 \times \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + 2\ln x-5 + c$ $= -\frac{1}{2} \ln x^2+4 - \tan^{-1}\left(\frac{x}{2}\right) + 2\ln x-5 + c$	Some students left out the middle term
12.	(c)(i) $\frac{1}{2} \frac{d}{dx} v^2 = -\frac{1}{50} (10-x)^3$ $v^2 = \frac{1}{100} (10-x)^4 + c$ $t=0, x=0, v=+10 \Rightarrow c=0 \text{ and } v^2 = \frac{1}{100} (10-x)^4, v > 0$ $\therefore v = \frac{1}{10} (10-x)^2$ Or $v \frac{dv}{dx} = \frac{-1}{50} (10-x)^3$ $\int v \cdot dv = \frac{-1}{50} \int (10-x)^3 \cdot dx$ $\left[\frac{v^2}{2} \right] = \frac{-1}{50} \left[\frac{(10-x)^4}{4 \times (-1)} \right] + c$ $\frac{v^2}{2} = \frac{1}{200} (10-x)^4 + c$ When $t=0, x=0, v^2 = 10$	Need to give justification for discarding the negative case

$$\frac{10^2}{2} = \frac{1}{200}(10-0)^4 + c$$

$$\frac{100}{2} = \frac{1}{200}10000 + c$$

$$50 = 50 + c$$

$$c = 0$$

$$\frac{v^2}{2} = \frac{1}{200}(10-x)^4 + 0$$

$$v^2 = \frac{1}{100}(10-x)^4$$

$$v = \pm \frac{1}{10}(10-x)^2$$

When $t=0$ moving right, $v>0$

$$v = \frac{1}{10}(10-x)^2$$

12. (c)(ii)

$$\frac{dx}{dt} = \frac{1}{10}(10-x)^2$$

$$\int \frac{1}{(10-x)^2} dx = \frac{1}{10} \int dt$$

$$\frac{1}{10-x} = \frac{1}{10}t + c$$

$$t=0, x=0 \Rightarrow c = \frac{1}{10}$$

$$\therefore \frac{10}{10-x} = 1+t$$

$$1 - \frac{x}{10} = \frac{1}{1+t}$$

$$\frac{x}{10} = 1 - \frac{1}{1+t}$$

$$x = \frac{10t}{1+t}$$

13.	(a) <u>To prove:</u> $\log_4 7$ is irrational Assume $\log_4 7$ is rational ① i.e. $\log_4 7 = \frac{p}{q}$ where $p, q \in \mathbb{Z}^+$ and p, q are relative primes $q \log_4 7 = p$ $\log_4 7^q = p$ $\therefore 7^q = 4^p$ ① LHS = 7^q $= 7 \times 7 \times 7 \times \dots$ i.e. product of q number of 7 with all factors are 7 RHS = 4^p $= 4 \times 4 \times 4 \times \dots$ i.e. product of p number of 4 with all factors are 4 Hence LHS $\neq$ RHS, which is a contradiction to the statement that $\log_4 7$ is rational ① $\therefore \log_4 7$ is irrational.	Mostly well done. (85% correct) Area of concern: (1) some students are not realising the contradiction is on the original statement. Hence their statement and conclusion are confusing.
13.	(b) <u>To prove:</u> $u_n = 3(2^{n+1}) - 1$ for all positive integers n given $u_0 = 5$, $u_n = 2u_{n-1} + 1$ for $n \geq 1$. <u>Proof:</u> Step 1: to prove true for $n = 1$. From closed definition: $u_1 = 3(2^{1+1}) - 1$ $= 3(2^2) - 1$ $= 12 - 1$ $\therefore u_1 = 11$ From recursive definition: $u_1 = 2u_{1-1} + 1$ $= 2u_0 + 1$ $= 2(5) + 1$ $\therefore u_1 = 11$ which agrees with the closed definition. Hence true for $n = 1$. Step 2: Assume true for $n = k$, $k \in \mathbb{Z}^+$, $u_k = 3(2^{k+1}) - 1$ RTP: when $n = k + 1$, $u_{k+1} = 3(2^{k+2}) - 1$ From recursive definition:: $u_{k+1} = 2u_{(k+1)-1} + 1$ $= 2u_k + 1$ $= 2[3(2^{k+1}) - 1] + 1$ (from assumption) $= 2 \times 3 \times 2^{k+1} - 2 + 1$	Poorly done. (50% correct) Area of concern: (1) some students finding the smallest n value as 0 whilst the proof is for $n \geq 0$. (2) some students do not realise how significant the assumption of $n = k$ is true for the proof of $n = k + 1$. (3) some students do not realise how significant the two different types of equations given and the fact that one of them is recursive.

	$= 3 \times (2 \times 2^{k+1}) - 1$ $\therefore u_{k+1} = 3(2^{k+2}) - 1 \text{ which agrees}$ with the closed definition. If true for $n = k$, hence proven true for $n = k + 1$. Step 3: Since proven true for $n = 1$, hence proven true for $n = 1 + 1 = 2$, $n = 2 + 1 = 3$, and so on. Therefore true for all positive integers n.	
13.	(c)(i) $z^5 = 1$ where $z = \cos \theta + i \sin \theta$, $-\pi \leq \theta \leq \pi$ $z^n = \cos n\theta + i \sin n\theta$ (De Moivre's theorem) $\cos 5\theta + i \sin 5\theta = 1$ Equating real terms: $\cos 5\theta = 1$ $5\theta = 2k\pi$, $k \in \mathbb{Z}$ $\therefore \theta = \frac{2k\pi}{5}$ where $k = 0, \pm 1, \pm 2$ $\therefore z_1 = \text{cis } 0 = \cos 0 + i \sin 0 = 1$ $z_2 = \text{cis } \frac{2\pi}{5}$ $z_3 = \text{cis } \frac{4\pi}{5}$ $z_4 = \text{cis } \left(-\frac{2\pi}{5}\right) = \bar{z}_2$	Mostly well done. (92.5% correct) Area of concern: (1) some students are not stating De Moivre's theorem when they used it.

13.	(c)(i)cont. $z_5 = \text{cis } \left(-\frac{4\pi}{5}\right) = \bar{z}_3$ Hence the roots of $z^5 = 1$ are $\text{cis } 0, \text{cis } \frac{2\pi}{5}, \text{cis } \left(-\frac{2\pi}{5}\right), \text{cis } \frac{4\pi}{5}, \text{cis } \left(-\frac{4\pi}{5}\right)$ (ii) $z^5 = 1$ $z^5 - 1 = 0$ $(z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5) = 0$ Since $z_4 = \bar{z}_2$ and $z_5 = \bar{z}_3$ $(z - 1)(z - z_2)(z - \bar{z}_2)(z - z_3)(z - \bar{z}_3) = 0$ $(z - 1)[z^2 - (z_2 + \bar{z}_2)z + z_2\bar{z}_2][z^2 - (z_3 + \bar{z}_3)z + z_3\bar{z}_3] = 0$ where $z_2 + \bar{z}_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \left(\cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}\right)$ $z_2 + \bar{z}_2 = 2 \cos \frac{2\pi}{5}$	Poorly done. (45% correct) Area of concern: (1) students are not showing how they obtain coefficient of z or the constant 1 of the quadratic factors.
-----	--	---

	Similarly, $z_3 + \bar{z}_3 = 2 \cos \frac{4\pi}{5}$ $z_2 \bar{z}_2 = \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right) \left(\cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5} \right)$ $= \cos^2 \frac{2\pi}{5} - i^2 \sin^2 \frac{2\pi}{5}$ $= \cos^2 \frac{2\pi}{5} - (-1) \sin^2 \frac{2\pi}{5}$ $= \cos^2 \frac{2\pi}{5} + \sin^2 \frac{2\pi}{5}$ $\therefore z_2 \bar{z}_2 = 1$ and similarly, $z_3 \bar{z}_3 = 1$ $\therefore (z-1) \left(z^2 - 2 \cos \frac{2\pi}{5} + 1 \right) \left(z^2 - 2 \cos \frac{4\pi}{5} + 1 \right) = 0$ $z^5 - 1 = (z-1) \left(z^2 - 2 \cos \frac{2\pi}{5} + 1 \right) \left(z^2 - 2 \cos \frac{4\pi}{5} + 1 \right)$ $(z-1) (z^4 + z^5 + z^2 + z + 1)$ $= (z-1) \left(z^2 - 2 \cos \frac{2\pi}{5} + 1 \right) \left(z^2 - 2 \cos \frac{4\pi}{5} + 1 \right)$ $\therefore z^4 + z^5 + z^2 + z + 1 = \left(z^2 - 2 \cos \frac{2\pi}{5} + 1 \right)$ $\times \left(z^2 - 2 \cos \frac{4\pi}{5} + 1 \right)$ (as required)	
--	--	--

13.	(c)(iii) $z^4 + z^5 + z^2 + z + 1 = \left(z^2 - 2 \cos \frac{2\pi}{5} + 1 \right)$ $\times \left(z^2 - 2 \cos \frac{4\pi}{5} + 1 \right)$ Substituting $z = 1$, $(1)^4 + (1)^5 + (1)^2 + (1) + 1 = \left((1)^2 - 2 \cos \frac{2\pi}{5} + 1 \right)$ $\times \left((1)^2 - 2 \cos \frac{4\pi}{5} + 1 \right)$ $5 = \left(2 - 2 \cos \frac{2\pi}{5} \right) \left(2 - 2 \cos \frac{4\pi}{5} \right)$ $5 = 2 \left(1 - \cos \frac{2\pi}{5} \right) \times 2 \left(1 - \cos \frac{4\pi}{5} \right)$ $\therefore 4 \left(1 - \cos \frac{2\pi}{5} \right) \left(1 - \cos \frac{4\pi}{5} \right) = 5 \quad (\text{as required})$	Mostly well done. (75% correct)
(iv)	$4 \left(1 - \cos \frac{2\pi}{5} \right) \left(1 - \cos \frac{4\pi}{5} \right) = 5$	Mostly well done. (62.5% correct) Area of concern:

	[Hint: $\cos \frac{4\pi}{5} = \cos 2\left(\frac{2\pi}{5}\right)$ i.e. double angle] $4\left(1 - \cos \frac{2\pi}{5}\right)\left(1 - \cos 2\left(\frac{2\pi}{5}\right)\right) = 5$ $4\left(1 - \cos \frac{2\pi}{5}\right)\left(1 - \left(1 - 2\sin^2\left(\frac{2\pi}{5}\right)\right)\right) = 5$ $4\left(1 - \cos \frac{2\pi}{5}\right) \times 2\sin^2\left(\frac{2\pi}{5}\right) = 5$ $8\left(1 - \cos \frac{2\pi}{5}\right)\sin^2\left(\frac{2\pi}{5}\right) = 5$ $8\left(1 - \cos \frac{2\pi}{5}\right)\left(1 - \cos^2 \frac{2\pi}{5}\right) = 5$ Let $x = \cos \frac{2\pi}{5}$, $8(1-x)(1-x^2) = 5$ $8(1-x-x^2+x^3) = 5$ $8x^3 - 8x^2 - 8x + 8 = 5$ $\therefore 8x^3 - 8x^2 - 8x + 3 = 0$ Hence $\cos \frac{2\pi}{5}$ is a root of the equation	(1) some students did not identify that $\cos \frac{4\pi}{5} = \cos 2\left(\frac{2\pi}{5}\right)$
--	--	--

14.	(a)(i) The direction vectors of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are $\begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$ respectively. $\begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} = 3 - 12 + 9$ $= 0$ $\therefore AB$ is perpendicular to CD. ① Alternative solution 1 (if they find E first and come back to this question): $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix}$ $\overrightarrow{CE} = \begin{pmatrix} 3-0 \\ 3-9 \\ 1-10 \end{pmatrix}$ $= \begin{pmatrix} 3 \\ -6 \\ -9 \end{pmatrix}$ $\overrightarrow{AB} \cdot \overrightarrow{CE} = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -6 \\ -9 \end{pmatrix}$ $= 9 - 36 + 27$ $= 0$ ①	Done well by most students
-----	--	-----------------------------------

$\therefore AB$ is perpendicular to CE , so AB is perpendicular to CD

since C , D and E are collinear.

$$(a)(ii) \quad \vec{r}_{AB} = \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -6 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 4 - 3\lambda \\ 5 - 6\lambda \\ 3\lambda \end{pmatrix}$$

$$\vec{r}_{CD} = \begin{pmatrix} 0 \\ 9 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} -\mu \\ 9 + 2\mu \\ 10 + 3\mu \end{pmatrix}$$

At the point of intersection

$$\begin{pmatrix} 4 - 3\lambda \\ 5 - 6\lambda \\ 3\lambda \end{pmatrix} = \begin{pmatrix} -\mu \\ 9 + 2\mu \\ 10 + 3\mu \end{pmatrix} \quad \textcircled{1}$$

$$\therefore 4 - 3\lambda = -\mu \quad (1)$$

$$5 - 6\lambda = 9 + 2\mu \quad (2)$$

$$3\lambda = 10 + 3\mu \quad (3)$$

$$(1) + (3): \quad 4 = 10 + 2\mu$$

14. (a)(ii) (1) + (3): $4 = 10 + 2\mu$

$$2\mu = -6$$

$$\mu = -3$$

$$\text{The point } E \text{ is } \begin{pmatrix} -(-3) \\ 9 + 2(-3) \\ 10 + 3(-3) \end{pmatrix} = (3, 3, 1). \quad \textcircled{1}$$

Note that there are many ways in which this solution can be set out - far too many to list. They all end with the same answer. One alternative solution is shown next page.

Alternative solution 1:

$$r_{AB} = \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 + \lambda \\ 5 + 2\lambda \\ -\lambda \end{pmatrix}$$

$$r_{CD} = \begin{pmatrix} 0 \\ 9 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} \mu \\ 9 - 2\mu \\ 10 - 3\mu \end{pmatrix}$$

At the point of intersection

$$\begin{pmatrix} 4 + \lambda \\ 5 + 2\lambda \\ -\lambda \end{pmatrix} = \begin{pmatrix} \mu \\ 9 - 2\mu \\ 10 - 3\mu \end{pmatrix}$$

$$\therefore 4 + \lambda = \mu \quad (1)$$

$$5 + 2\lambda = 9 - 2\mu \quad (2)$$

$$-\lambda = 10 - 3\mu \quad (3)$$

$$\text{sub (1) in (3): } -\lambda = 10 - 3(4 + \lambda)$$

$$= 10 - 12 - 3\lambda$$

$$2\lambda = -2$$

$$\lambda = -1$$

$$\text{The point } E \text{ is } \begin{pmatrix} 4 + (-1) \\ 5 + 2(-1) \\ -(-1) \end{pmatrix} = (3, 3, 1).$$

①

①

14.	(b) Write $z = 1 - \sqrt{3}i$ in polar form $r(\cos\theta + i\sin\theta)$ $r = \sqrt{1+3} = \sqrt{4} = 2$ $\tan\theta = \frac{-\sqrt{3}}{1} = -\sqrt{3}, \theta = \frac{-\pi}{3}$ $z = 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$ $\gamma = (2)^{\frac{1}{2}}\left(\cos\frac{\left(-\frac{\pi}{3} + 2k\pi\right)}{2} + i\sin\frac{\left(-\frac{\pi}{3} + 2k\pi\right)}{2}\right),$ $k = 0, 1$	

14.	(b) continues For $k = 0$, $\gamma = \sqrt{2}\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$ $= \sqrt{2}\left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)$ $= \frac{\sqrt{6} - \sqrt{2}i}{2}$ For $k = 1$, $\gamma = \sqrt{2}\left(\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right)$ $= \sqrt{2}\left(-\frac{\sqrt{3}}{2} + \frac{i}{2}\right) = \frac{-\sqrt{6} + \sqrt{2}i}{2}$ $= \frac{-\sqrt{6} + \sqrt{2}i}{2}$ The square roots are $\frac{\sqrt{6} - \sqrt{2}i}{2}$ and $\frac{-\sqrt{6} + \sqrt{2}i}{2}$ $\frac{\sqrt{6}}{2} - \frac{\sqrt{2}i}{2}, \quad \frac{-\sqrt{6}}{2} + \frac{\sqrt{2}i}{2}$	
-----	---	--

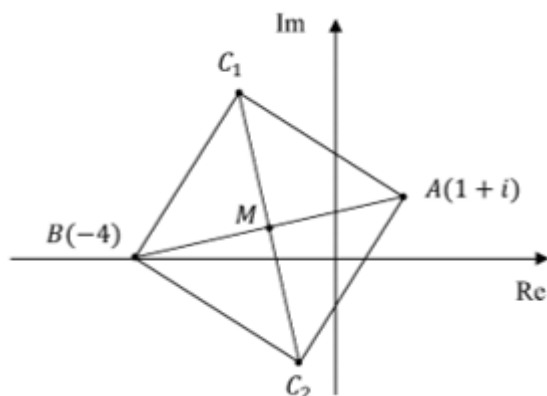
14.	(c) $\omega = \cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ $\omega^2 + \frac{1}{\omega^2} = \omega^2 + \overline{\omega^2} \quad \text{since } \omega^2 = \omega ^2 = 1 ^2 = 1$ $= 2\operatorname{Re}(\omega^2)$ $= 2\operatorname{Re}\left(\cos\frac{4\pi}{5} + i\sin\frac{4\pi}{5}\right) \quad \text{using DMT}$ $= 2\cos\frac{4\pi}{5} \quad \textcircled{1}$ $\therefore \left(2\cos\frac{4\pi}{5}\right)^2 + 2\cos\frac{4\pi}{5} - 1 = 0$ $4\cos^2\frac{4\pi}{5} + 2\cos\frac{4\pi}{5} - 1 = 0$ $\cos\frac{4\pi}{5} = \frac{-2 \pm \sqrt{2^2 - 4(4)(-1)}}{2(4)}$ $= \frac{-2 \pm \sqrt{20}}{8}$ $= \frac{-2 \pm 2\sqrt{5}}{8}$	
-----	--	--

①

Must justify why the positive case is discarded

①

14. (d)(i)



From the diagram we can see that C can lie in the second or third quadrants.

The two possible right-angled triangles form a square centred at the midpoint of AB , which is marked as M . Since the diagonals of a square bisect at right-angles, we rotate MA 90° anticlockwise or clockwise about M to find the possible positions for C .

To find the two possible values for z_C , we start at z_M and rotate $z_A - z_M$ 90° anticlockwise (multiply by i) or clockwise (multiply by $-i$).

$$\begin{aligned} z_M &= \frac{1+i-4}{2} \\ &= -\frac{3}{2} + \frac{i}{2} \end{aligned}$$

$$z_A - z_M = 1+i - \left(-\frac{3}{2} + \frac{i}{2}\right)$$

Common alternative methods were to

- Use dot product since right angled at C

$$\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}\right) \cdot \left(\begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}\right) = 0 \quad \text{or}$$

- Pythagoras's Theorem

$$\begin{aligned} |AC|^2 + |BC|^2 &= |AB|^2 \\ \sqrt{(x-1)^2 + (y-1)^2} + \sqrt{(x+4)^2 + (y-0)^2} &= \sqrt{(1-4)^2 + (1-0)^2} \end{aligned}$$

to find

$$\left(x + \frac{3}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{26}{4}$$

and then use the fact that ABC is an isosceles triangle so $AC=BC$ and

①

(d)(ii) The argument of the vector from B to any point Z on the diagonal, representing the complex number z, is always π radians greater than the argument from A to Z.	$z - z_A = z - z_B$ $x + iy - (1 + i) = x + iy - (-4 + 0i)$ $\sqrt{(x-1)^2 + (y-1)^2} = \sqrt{(x+4)^2 + y^2}$ To obtain $y = -5x - 7$ and substitute into equation of circle above to find $x = -2, y = 3$ or $x = -1, y = -2$
--	--

14. (d)(ii) continues

So, $\arg(z - z_B) - \arg(z - z_A) = \pi$
 $\therefore \arg(z - (-4 + 0i)) - \arg(z - (1 + i)) = \pi$
 $\therefore \arg(z + 4 + 0i) - \arg(z - 1 - i) = \pi$

$\therefore x_1 = 4, y_1 = 0, x_2 = -1, y_2 = -1, \theta = \pi.$ ①

Alternatively
 The argument of the vector from A to any point Z on the diagonal, representing the complex number z , is always π radians less than the argument from B to Z. ①

$\arg(z - z_A) - \arg(z - z_B) = -\pi$
 $\therefore \arg(z - (1 + i)) - \arg(z - (-4 + 0i)) = -\pi$
 $\therefore \arg(z - 1 - i) - \arg(z + 4 + 0i) = -\pi$

①

15. (a)(i) Given: When $t = 0$, $x = x_{\max}$, i.e. $v = 0$. ie.

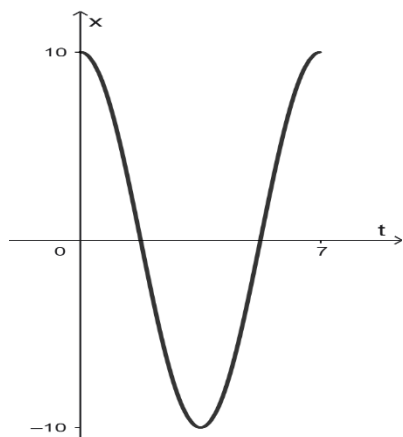
$$x = a \cos nt + c$$

Amplitude = 10, period = 7 s

$$\frac{2\pi}{n} = 7$$

$$n = \frac{2\pi}{7}$$

Let the equilibrium position be at O ,



Hence the position of particle at time t s is

$$x = 10 \cos \frac{2\pi}{7} t \quad \text{①}$$

Mostly well done. (7.5% correct)

15. (a)(ii) When $x = 5$, $10 \cos \frac{2\pi}{7} t = 5$

$$\cos \frac{2\pi}{7} t = \frac{1}{2}$$

$$\frac{2\pi}{7} t = \frac{\pi}{3}$$

$$\frac{2}{7} t = \frac{1}{3}$$

$$t = \frac{7}{6} \text{ s}$$

$\therefore$ the particle takes $1\frac{1}{6}$ seconds

①

Mostly well done. (55% correct)

Area of concern:

(1) careless errors caused many wrong answers.

15. (i) $a = -(1+v)$

$$v \frac{dv}{dx} = -(1+v)$$

$$\frac{dv}{dx} = -\frac{1+v}{v}$$

$$\int_u^v \frac{v}{1+v} dv = - \int_0^x 1 dx$$

$$\int_u^v \frac{v+1-1}{1+v} dv = - \int_0^x 1 dx$$

Mostly well done. (55% correct)

Area of concern:

(1) careless errors caused many wrong answers.

(2) few students forgot that the initial velocity is $u \text{ ms}^{-1}$, and wrote 0 which makes the subsequent working so much easier.

	$\int_u^v \frac{(v+1)}{1+v} - \frac{1}{1+v} dv = - \int_0^x 1 dx$ $\int_u^v 1 - \frac{1}{1+v} dv = - \int_0^x 1 dx$ $\left[v - \ln 1+v \right]_u^v = - \left[x \right]_0^x$ $\left[v - \ln 1+v \right] - \left[u - \ln 1+u \right] = - \left[x - 0 \right]$ $v - \ln 1+v - u + \ln 1+u = -x$ $v - u - (\ln 1+v - \ln 1+u) = -x$ $v - u - \left(\ln \frac{ 1+v }{1+u} \right) = -x$	
--	---	--

15.	(b)(i)cont. $\therefore x = u - v + \ln \frac{ 1+v }{1+u}$ $a = -(1+v)$ $\frac{dv}{dt} = -(1+v)$ $\int_u^v \frac{1}{1+v} dv = - \int_0^t dt$ $\left[\ln 1+v \right]_u^v = - \left[t \right]_0^t$ $\ln 1+v - \ln 1+u = -t$ $\ln \frac{ 1+v }{1+u} = -t$ $\frac{ 1+v }{1+u} = e^{-t}$ $ 1+v = (1+u)e^{-t} \text{ since initial } v > 0$ $1+v = (1+u)e^{-t}$ $\therefore v = (1+u)e^{-t} - 1$ $v = (1+u)e^{-t} - 1$ $\frac{dx}{dt} = (1+u)e^{-t} - 1$ $\int_0^x dx = \int_0^t (1+u)e^{-t} - 1 dt$ $\left[x \right]_0^x = \left[-(1+u)e^{-t} - t \right]_0^t$ $x = \left[-(1+u)e^{-t} - t \right] - \left[-(1+u)e^{-0} - 0 \right]$ $x = -(1+u)e^{-t} - t + (1+u)$	
-----	--	--

	$\therefore x = u - (1+u)e^{-t} + 1 - t$ (b)(ii) From part (ii) $v = (1+u)e^{-t} - 1$ $(1+u)e^{-t} = v + 1 \quad \text{--- ①}$ $x = u - (1+u)e^{-t} + 1 - t \quad \text{--- ②}$ Sub ① into ② $x = u - (v+1) + 1 - t$ $x = u - v - 1 + 1 - t$ $x = u - v - t$ $\therefore x + v + t = u \quad (\text{as required})$ (b)(iii) At rest, $v = 0$, $x = u - (0) + \ln \frac{1+(0)}{1+u}$ [from (ii)]	Poorly done. (50% correct) Area of concern: (1) students did not know how to work this out. Poorly done. (40% correct) Area of concern: (1) students did not know how to work this out.
--	--	---

15.	(b)(iv)cont. $x = u + \ln \frac{1}{1+u}$ $x = u + \ln(1+u)^{-1}$ $\therefore x = u - \ln(1+u)$ m i.e. the distance the particle travelled from O before coming to rest. At rest, $v = 0$, $x = u - \ln(1+u)$, $x + v + t = u \quad [\text{from (iii)}]$ $[u - \ln(1+u)] + (0) + t = u$ $[u - \ln(1+u)] + (0) + t = u$ $-\ln(1+u) + t = 0$ $\therefore t = \ln(1+u) \text{ s i.e. the time taken to come to rest.}$	
15.	(c) $\int_0^1 \sin^{-1} x \, dx = [uv]_0^1 - \int_0^1 u'v \, dx$ where $u = \sin^{-1} x$ $u' = \frac{1}{\sqrt{1-x^2}}$ $v' = 1$ $v = x$ $= [x \sin^{-1} x]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx$	Mostly done. (62.5% correct) Area of concern: (1) students through careless integration obtain answer $\frac{\pi}{2} + 1$ rather than $\frac{\pi}{2} - 1$. (27.5%)

	$= \left[(1) \sin^{-1}(1) - 0 \right] - \left(-\frac{1}{2} \right) \int_0^1 \frac{-2 \times x}{\sqrt{1-x^2}} dx$ $= \frac{\pi}{2} + \frac{1}{2} \int_0^1 -2x(1-x^2)^{-\frac{1}{2}} dx$ $= \frac{\pi}{2} + \frac{1}{2} \left[\frac{2(1-x^2)^{\frac{1}{2}}}{1} \right]_0^1$ $= \frac{\pi}{2} + \left[\sqrt{1-x^2} \right]_0^1$ $= \frac{\pi}{2} + \left[\sqrt{1-(1)^2} - \sqrt{1-(0)^2} \right]$ $= \frac{\pi}{2} + [0-1]$ $\therefore \int_0^1 \sin^{-1} x \, dx = \frac{\pi}{2} - 1$	
--	---	--

16.	(a) $\frac{a}{b} \geq \frac{c}{a}$ $\Rightarrow a^2 \geq b$: $\Rightarrow 1 \cdot a^2 \geq 1 \cdot b$: $\Rightarrow 21 \cdot a \geq 1 \cdot b + 1 \cdot a$ $\Rightarrow 1 \cdot a \geq \frac{1 \cdot b + 1 \cdot a}{2}$. Now $(x+y)^2 = (x-y)^2 + 4xy \geq 4xy \Rightarrow (x+y)^2 \geq 4xy$. Now for $x \geq 0$ and $y \geq 0$ $\frac{x+y}{2} \geq \sqrt{xy}$. Substitute $x = \ln b$ and $y = \ln c$ gives $\frac{\ln b + \ln c}{2} \geq \sqrt{\ln b \ln c}.$ Thus $\ln a \geq \sqrt{\ln b \ln c}$. i.e. $(\ln a)^2 \geq \ln b \ln c$ $\frac{\ln a}{\ln b} \geq \frac{\ln c}{\ln a}.$	Many students over simplified the process and/or made errors with $\ln bc \neq \ln b \ln c$ Or taking the log of both sides of $\frac{a}{b} \geq \frac{c}{a}$ gives $\ln \frac{a}{b} \geq \ln \frac{c}{a}$ NOT $\frac{\ln a}{\ln b} \geq \frac{\ln c}{\ln a}$
16.	(b)(i) $I_n = \int_1^e (\ln x)^n dx, n = 0, 1, 2, \dots$ $I_{n+1} = \int_1^e 1 \cdot (\ln x)^{n+1} dx$	

	$I_0 = \int_1^e 1 \, dx = e - 1$ $I_1 = e - (0 + 1) I_0$ $= e - (e - 1)$ $= 1$ $I_2 = e - (1 + 1) I_1$ $= e - 2$	Need proof of result, not just substitution
		$I_2 = e - (1 + 1) I_1 \quad \text{not} \quad I_2 = e - (2 + 1) I_1$

16.	(b)(ii) $I_n = (-1)^{n+1} n! + (-1)^n e n! \sum_{r=2}^n \frac{(-1)^r}{r!}$ Step 1: When $n = 2$, LHS = I_2 [from part (i)] $= e - 2$ RHS = $(-1)^{2+1} 2! + (-1)^2 e \times 2! \left(\frac{1}{2!} \right)$ $= -2! + e$ $= e - 2$ $\therefore$ true for $n = 2$. Step 2: Assume true for $n = k$, $I_k = (-1)^{k+1} k! + (-1)^k e k! \sum_{r=2}^k \frac{(-1)^r}{r!}$ <u>RTP</u>: When $n = k + 1$, $I_{k+1} = (-1)^{k+2} (k+1)! + (-1)^{k+1} e (k+1)! \sum_{r=2}^{k+1} \frac{(-1)^r}{(r+1)!}$ From part (i), $I_{k+1} = e - (k+1) I_k$ $= e - (k+1) \left\{ (-1)^{k+1} k! + (-1)^k e k! \sum_{r=2}^k \frac{(-1)^r}{r!} \right\}$ (from assumption) $= e + (-1)^{k+2} (k+1)! + (-1)^{k+1} e (k+1)! \sum_{r=2}^k \frac{(-1)^r}{r!}$ $= (-1)^{k+2} (k+1)! + (-1)^{k+1} e (k+1)! \left\{ \sum_{r=2}^k \frac{(-1)^r}{r!} + \frac{1}{(-1)^{k+1} (k+1)!} \right\}$ $= (-1)^{k+2} (k+1)! + (-1)^{k+1} e (k+1)! \left\{ \sum_{r=2}^k \frac{(-1)^r}{r!} + \frac{(-1)^{k+1}}{(k+1)!} \right\}$
------------	---

	If true for $n = k$, hence proven true for $n = k + 1$. Step 3: Since proven true for $n = 2$, $n = 2 + 1 = 3$, and so on, hence true for all positive integers of $n \geq 2$.	Many students struggled to prove true for $n=k+1$
16.	(c)(i) $\vec{a}, \vec{b}, \vec{c}$ form a triangle $\Rightarrow \vec{a} + \vec{b} + \vec{c} = \vec{0}$ $\begin{aligned}\vec{u} + \vec{v} + \vec{w} &= (\vec{b} \cdot \vec{c})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b} + (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} + (\vec{b} \cdot \vec{a})\vec{c} - (\vec{c} \cdot \vec{b})\vec{a} \\ &= (\vec{b} \cdot \vec{c})\vec{a} - (\vec{c} \cdot \vec{b})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b} + (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} + (\vec{b} \cdot \vec{a})\vec{c} \\ &= \vec{0} \quad \text{since } \vec{p} \cdot \vec{q} = \vec{q} \cdot \vec{p}\end{aligned}$ Hence $\vec{u}, \vec{v}, \vec{w}$ also form a triangle	Many tried to expand terms such as $(\vec{b} \cdot \vec{c})\vec{a}$ incorrectly NOT the same as $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ Some did not recognise $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ as the condition for forming a triangle

16.	(c)(ii) $\vec{u} \cdot \vec{c} = ((\vec{b} \cdot \vec{c})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b}) \cdot \vec{c}$ $\begin{aligned}&= (\vec{b} \cdot \vec{c})\vec{a} \cdot \vec{c} - (\vec{c} \cdot \vec{a})\vec{b} \cdot \vec{c} \\ &= 0\end{aligned}$ (c)(iii) From (ii), $\vec{u}$ and $\vec{c}$ are perpendicular. Similarly, $\vec{v}$ and $\vec{a}$, and $\vec{w}$ and $\vec{b}$ are respectively perpendicular. The sides of the triangle formed with $\vec{u}, \vec{v}, \vec{w}$ are perpendicular to the sides of the triangle formed with $\vec{a}, \vec{b}, \vec{c}$, hence the two triangles are equiangular, that is are similar.	Done well by most Some tried to expand incorrectly $\vec{u} \cdot \vec{c} = 0$, so $\vec{u}$ and $\vec{c}$ are perpendicular Similarly $\vec{v} \cdot \vec{a} = 0$ and $\vec{w} \cdot \vec{b} = 0$
-----	---	---